

DW01_mux_any

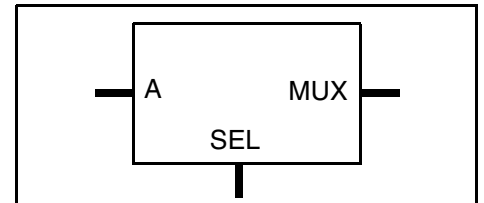
Universal Multiplexer

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word lengths
- Saves coding time by eliminating the need to code multiplexers (MUX's) explicitly
- Increases design abstraction
- Uses 8-to-1 MUX's where possible

Revision History



Description

DW01_mux_any is an universal multiplexer. This component selects a subrange of the input *A*, and places it on the output *MUX*. The select input, *SEL*, multiplexes the subrange of *A* to *MUX*. If necessary, *A* is padded with high order bits of zero. The widths of the inputs and outputs are fixed by the parameters *A_width*, *SEL_width*, and *MUX_width*.

DW01_mux_any uses 8-to-1 MUX's whenever possible to get the fastest possible implementation with the lowest gate count.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>A_width</i>	Input	Data input bus
SEL	<i>SEL_width</i>	Input	Select input
MUX	<i>MUX_width</i>	Output	Multiplexed data out

Table 1-2 Parameter Description

Parameter	Values	Description
<i>A_width</i>	≥ 1	Word length of <i>A</i>
<i>SEL_width</i>	≥ 1	Word length of <i>SEL</i>
<i>MUX_width</i>	≥ 1	$A((SEL + 1) \times MUX_width - 1 \text{ downto } SEL * MUX_width)$
<i>bal_str</i> ^a	0 or 1 Default: 0	Controls the symmetric structure of the 'str' implementation. 1 = Use symmetric structure with 'str' implementation.

a. Prior to release 2007.12-SP2, this parameter is ignored.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW01.DW01_MUX_ANY_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW01_mux_any_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW01_mux_any.v	Verilog simulation model source code

Related Topics

- [Logic – Combinational Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_mux_any_inst is
  generic (inst_A_width : POSITIVE := 8;
           inst_SEL_width : POSITIVE := 8;
           inst_MUX_width : POSITIVE := 8);
  port (inst_A : in std_logic_vector(inst_A_width-1 downto 0);
        inst_SEL : in std_logic_vector(inst_SEL_width-1 downto 0);
        MUX_inst : out std_logic_vector(inst_MUX_width-1 downto 0));
end DW01_mux_any_inst;

architecture inst of DW01_mux_any_inst is
begin
  -- Instance of DW01_mux_any
  U1 : DW01_mux_any
    generic map ( A_width => inst_A_width, SEL_width => inst_SEL_width,
                  MUX_width => inst_MUX_width )
    port map ( A => inst_A, SEL => inst_SEL, MUX => MUX_inst );
end inst;

-- pragma translate_off
configuration DW01_mux_any_inst_cfg_inst of DW01_mux_any_inst is
  for inst
  end for; -- inst
end DW01_mux_any_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW01_mux_any_inst( inst_A, inst_SEL, MUX_inst );

    parameter A_width = 8;
    parameter SEL_width = 8;
    parameter MUX_width = 8;

    input [A_width-1 : 0] inst_A;
    input [SEL_width-1 : 0] inst_SEL;
    output [MUX_width-1 : 0] MUX_inst;

    // Instance of DW01_mux_any
    DW01_mux_any #(A_width, SEL_width, MUX_width)
        U1 ( .A(inst_A), .SEL(inst_SEL), .MUX(MUX_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 3 ■ Added this Revision History table and the document links on this page

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